
EE1060: Differential Equations and Transform Techniques

Lecture 13: General Solutions

Topics :

1. Linear independence and Wronskian
 2. General Solutions of Homogeneous and nonhomogeneous ODE
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Linear Independence, fundamental set of Solutions and Property of Wronskian

For the ODE $L[y]=0$, the linearly independent set $\{y_1, \dots, y_n\}$ is called the fundamental set of solutions.
 $\downarrow$
 Wronskian.

Property of Wronskian: $w(y_1, \dots, y_n)$ is either $\equiv 0$ on the I or it's non-zero on I .

Proof :- ($n=2$):

$$w = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

y_1, y_2 are solutions to

$$y'' + p(x)y' + q(x)y = 0$$

$$\Rightarrow y'' = -[p(x)y' + q(x)y] \quad \hookrightarrow \textcircled{2}$$

$$w' = y_1 y_2'' + y_1' y_2' - y_2 y_1'' - y_2' y_1'$$

$$w' = y_1 y_2'' - y_2 y_1'' \rightarrow \textcircled{1}$$

from $\textcircled{2}$

$$w' = -y_1 [p(x)y_2' + q(x)y_2] + y_2 [p(x)y_1' + q(x)y_1]$$

$$= p(x) [y_2 y_1' - y_1 y_2'] = -p(x)w$$

$$w' + p(x)w = 0 \Rightarrow \frac{1}{w} \frac{dw}{dx} = -p(x)$$

$$w = c e^{-\int p(x) dx} \Rightarrow \begin{cases} \text{if } c=0, w \equiv 0 \\ \text{if } c \neq 0, w \neq 0 \text{ on } I. \end{cases}$$

Linear Independence and Wronskian

If $W(y_1, y_2)(x_0) \neq 0$ for some x_0 in I , then y_1, y_2 are linearly independent.
 (Solutions for $y'' + p(x)y' + Q(x)y = 0$)

Proof by contradiction:- y_1, y_2 are linearly dependent and $W \neq 0$ at x_0
 $\Rightarrow W \neq 0$ on I .

$$\exists c_1, c_2 \neq 0 : c_1 y_1 + c_2 y_2 = 0 \rightarrow (1)$$

$$\frac{d}{dx} (1) \Rightarrow c_1 y_1' + c_2 y_2' = 0 \rightarrow (2)$$

$$\underbrace{\begin{bmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{bmatrix}}_A \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{has a non-trivial sol} \\ \text{or } \vec{c} \neq 0.$$

$\Rightarrow A$ is singular $\Rightarrow |A| = 0$ which is a contradiction

Uniqueness of Solutions to IVP and Existence of Fundamental Set

For $\{ L[y] = g(x) \text{ where } a_n(x), \dots, a_0(x), g(x) \text{ are continuous on } I \text{ and } y^{(i)}(x_0) = y_i \}$
 the IVP
 the solution exists & is unique.

$$\begin{bmatrix} y(x_0) \\ \vdots \\ y^{(n)}(x_0) \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \sum y_i \vec{e}_i$$

Existence of fundamental set of solutions:- For $L[y] = 0$, $\exists$ a fundamental set of solutions $\{y_1(x), \dots, y_n(x)\}$

Proof for $n=2$:- $y_1(x)$ and $y_2(x)$ are solutions for the ODE $y'' + p(x)y' + q(x)y = 0$

y_1 is a solution: $y'' + p(x)y' + q(x)y = 0$ with IC $y(x_0) = 1$, and $y'(x_0) = 0$

y_2 " " : $y'' + p(x)y' + q(x)y = 0$ " " $y(x_0) = 0$ and $y'(x_0) = 1$

y_1 & y_2 are linearly independent:

$$\begin{vmatrix} y_1(x_0) & y_2(x_0) \\ y_1'(x_0) & y_2'(x_0) \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \neq 0$$

$$\Rightarrow W(x_0) \neq 0 \Rightarrow W \neq 0$$

$\Rightarrow y_1$ & y_2 are linearly independent

General Solution of Homogeneous IVP

every solution of $L[y]=0$ can be expressed as a linear combination of the fundamental set of solutions.
 $\hookrightarrow$ with given IC.

proof :- ($n=2$) : y_1 & y_2 are fundamental solutions for $y'' + p(x)y' + q(x)y = 0 \rightarrow \textcircled{1}$

$y = c_1 y_1(x) + c_2 y_2(x)$ is also a solution for $\textcircled{1}$

$\phi(x)$ is a solution $L[y]=0$ with IC $y(x^0) = y_0$ and $y'(x^0) = y_1$
 $= \phi(x^0) \quad y'(x^0) = \phi'(x^0) \} \rightarrow \textcircled{2}$

Construct y_1 & y_2 such that $y_1 : \textcircled{1}$ with IC $y(x^0) = 1$ and $y'(x^0) = 0$

$y_2 : \textcircled{2}$ " " $y(x^0) = 0$ and $y'(x^0) = 1$

$y = y_0 y_1(x) + y_1 y_2(x)$ is a sol to $\textcircled{2}$

by uniqueness $y(x) \equiv \phi(x)$